TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Mark Courage	Email	mark.courage@tno.nl
Institute Name	TNO Nederlandse Organisatie voor toegepast-natuurwetenschappelijk onderzoek TNO (Netherlands Organisation for applied scientific research TNO)			
Address	Anna van Buerenplein 1, 2595 DA The Hague, The Netherlands			

2. Technology Overview

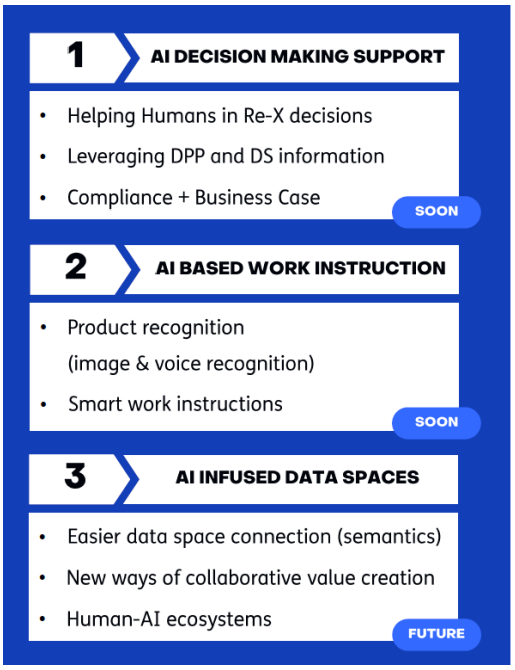
Title of Proposed Technology	**AI-enabled Decision Support for Re-X Activities**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical & materials		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Clean Production	*Process Improvement Technologies*	*R&D / Engineering Services*	*Other R&D / Engineering-Related Technologies*
Concept of Proposed Technology	As industries move toward circularity, the decision to **Repair, Reuse, Remanufacture, or Recycle (Re-X)** becomes a massive data challenge. Unlike new manufacturing, incoming products for Re-X are complex and "non-standard" because their condition depends on their unique usage history. Process engineers and operations cannot manually cross-reference the expanding multi-modal information for every product, while ensuring that the cost of human reasoning labor does not exceed the value of the component itself. At the same time, utilizing all the different potential information sources is required, as each one can bring unique relevant information that is needed for correct process decision. Information can come from, for example: Digital Product Passports (DPP), technical tolerances, economic trade-offs, company goals, and carbon footprint goals for every single part. Additionally, to take a decision about a product next step, a deep understanding is required into the specifics of a product, the corresponding rules that guide a choice. This information is currently often not			

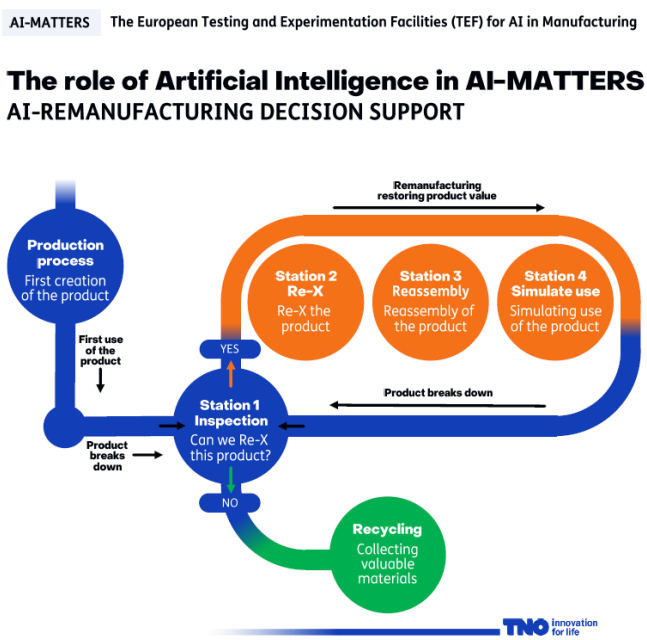
semantically recorded and/or standardized, where expertise is locked in the minds of seasoned professionals, making it difficult to transfer effectively which can in turn lead to loss in case of leave of retirement [Nikolova et al., 2025b; Stolwijk et al., 2024].
Because technical manuals and standard operating procedures are highly detailed and complex, it takes new employees a long training period to become fully operational. This creates a bottleneck in onboarding and slows down operational responsiveness.

To optimize those production processes, support the handling of the multi-modal comprehensive input of product data, and enhance the human operation, a Re-X Decision support system is proposed, that aims to automate support workflows for Re-X choices of products. It uses different sources of information together with systematized knowledge to determine the most sustainable and cost-effective Re-X step in minimal time. Decisions are routed to an employee to provide an actionable guidance: support on scheduling inspections, repair actions, setup changes or process adjustments.

Such a system would combine two types of reasoning (symbolic and sub-symbolic) to create Hybrid Neuro-Symbolic AI solution [Van Bekkum et al., 2021; De Boer et al., 2025]. Products, processes, machines, and constraints are defined as interoperable concepts and relations, enabling consistent reasoning across product structures, lifecycle metadata, and shopfloor instructions. The result is a closed-loop pipeline: aligned data → formalized rules → context-aware reasoning → actionable instructions.

The potential of this work is beyond a single product or process, as it would not only enhance the product operations in the case at hand, but also provide a methodological and deeper systematized understanding of the underlying manufacturing processes and help systematize knowledge in a machine-readable way for similar cases and other products.





Rationale for Joint R&D Project

The development of a **Re-X Decision Support System** is a multidisciplinary challenge that requires a synergy between theoretical innovation and real-world industrial applications. Collaboration is essential to bridge the gap between current State-of-the-Art (SoTA) research and the operational requirements of modern manufacturing.

- **Intersection of domains:** Successful implementation requires the fusion of "tribal knowledge" with advanced computational logic. An **SME partner** is critical to provide the real-world production data and the heuristic "brain-knowledge" (the *how* and *why* behind current manual decisions). **TNO** brings a proven track record in translating such complex human expertise into

	hybrid reasoning algorithms and structured knowledge systems [[Burghouts et al., 2025],[Hillerström et al., 2025],[Holvoet et al., 2025]]. This builds directly upon TNO's previous experience in high-tech manufacturing and agent-based systems [Sidorenko et al., 2023], as demonstrated in projects like **MAS4AI** and **NXTGEN Hightech**, and foundational research into neuro-symbolic design patterns [[Van Bekkum et al., 2021],[De Boer et al., 2025]] and AAS-based agent modelling [Nikolova et al., 2024]. • **Standardization of Semantic Models:** From a research perspective, TNO provides the methodology for creating specialized vocabularies and taxonomies that define information ecosystem, containing different sources of information, amongst which the **Digital Product Passport (DPP)** [Nikolova et al., 2025a, [Nikolova and Rongen, 2023],[Van Leeuwen et al., 2022], [Bouter et al., 2021]]. However, for these models to be actionable, they must be co-developed with an SME to ensure interoperability with actual production data, such as Bill of Materials (BOM) and inspection logs. This collaboration ensures that the resulting semantic structures are not only scientifically sound but industrially robust. • **Push for circularity:** Striving for a **Circular Economy** is a macro-scale goal that exceeds the capacity of any single stakeholder. It requires a "meet-in-the-middle" approach where research institutions provide the technological "push" and manufacturers provide the "market-pull." By collaborating, the partners can move beyond the current limitations—where components are often discarded due to the high labor costs of manual inspection—and move toward an **Autonomous Factory** model where Re-X decisions are automated, traceable, and economically viable **[Nikolova et al., 2025b]**.
Key Words	1. Re-X 2. Remanufacturing, Repair, Reuse decision support 3. Decision-Making 4. Artificial Intelligence 5. Knowledge Representation & Reasoning

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (6-7)
R&D Objectives	*Knowledge Level:* • **Semantic Representation of Relevant Knowledge:** Development and enrichment of domain-specific vocabularies (taxonomies and ontologies) for concrete process and product types (e.g., semiconductor products, semiconductor manufacturing processes, product specifications). • **Machine-Readable Formalization of Decision Logic:** Transformation of expert decision-making processes into structured, machine-readable rules, including choice criteria, parameter dependencies, and specification requirements tailored to concrete operational scenarios. • **Integration of Multimodal Product Information:** Methods to ingest, interpret, and unify heterogeneous incoming data streams (e.g., Digital Product Passports with lifecycle information, visual inspection results, sensor data, datasheets, and other documentation). • **Lifecycle Product Information:** Development of methods and data models to capture, structure, and maintain product- and process-level information within a Digital Product Passport, enabling standardized access to lifecycle data, traceability, and interoperability across remanufacturing steps. *Reasoning Level:* • **Hybrid AI Decision-Making Engine:** Development of a reasoning algorithm combining symbolic/semantic AI with data-driven approaches to enable robust, context-aware decision-making based on the enriched knowledge base. • **Traceable and Trustworthy Reasoning Framework:** Implementation of transparent reasoning mechanisms that generate verifiable logs and explanations, supporting validation, verification, and auditability of decisions. • **Human–Machine Interaction (HMI) Interface:** A user-facing application enabling operators and engineers to interact with the AI reasoner, explore decision rationales, adjust inputs, and access detailed knowledge representations.			

R&D Target Specifications	*Knowledge Level:*
	• Knowledge graphs developed/enriched for product and process specifications:
	  ○ *Minimum* 2 domain ontologies (product + process) developed or enriched
	  ○ *Coverage*: >70% of key product attributes and >70% of critical process steps represented in the knowledge graph
	• Rules defined for decision process and specification:
	  ○ 40 machine-readable decision rules formalized (parameter limits, routing choices, inspection thresholds…)
	  ○ 100% traceability of rule provenance (source expert, process document, data source)
	• Digital Product Passport generated and maintained automatically:
	  ○ Generation of a DPP instance with >70% structured data completeness
	  ○ Automatic ingestion from at least 2 heterogeneous data sources (e.g., inspection images, datasheets, lifecycle metadata)
	• Multimodal information integration pipeline operational:
	  ○ Support for at least 3 data modalities (vision, text, lifecycle metadata)
	
	Reasoning Level:
	• Decisions taken by Hybrid AI algorithm: >20% increase of decision accuracy; >50% of decisions automated			

R&D Plan	• Transparency and validation of reasoning: >80% rule correctness via human validation • Number of decisions automated by system: Scalable framework capable of x decisions per hour The R&D developments in this project will build on existing expertise TNO has developed in other projects. Components with starting ~TRL3-4 will be used as a basis, targeting novel systems at TRL7.
	Phase 1: Scoping, Requirements & Detailed R&D Planning (Month 1-2) • *T1. Process Definition & Analysis Map* the relevant remanufacturing process, identify decision points, required parameters, and constraints. • *T2. Requirements & Information Sources* Define technical requirements for Knowledge Graphs, Digital Product Passport data, multimodal inputs, and reasoning needs. • *T3. Data & Knowledge Modelling Plan* Draft modelling approach (RDF/OWL, rule types, expected entities and relations). • *T4. Detailed Project Plan* for Phase 2 Finalize scope, datasets, metrics, and timeline based on findings. • *T5. Architecture* for use case setup Draft the components and systems interactions in a use case setup. *Milestones: (M1)* Validated process and decision-point mapping *(M2)* Specification of required product/process knowledge & multimodal data *(M3)* Detailed Phase 2 R&D plan approved **Phase 2: Development, Integration, and Validation** • *T5. Formalize Knowledge (RDF / Knowledge Graph)* Encode product and process specifications; develop taxonomies and rules; align with DPP structure. • *T6. Gather and Integrate Multimodal Information* Connect visual inspection data, lifecycle data (DPP), datasheets, and other structured/unstructured inputs. • *T7. Develop Hybrid AI Core* Implement reasoning engine combining KG-based logic with data-driven components. • *T8. Connect Information & Knowledge to Reasoner* Build pipelines linking multimodal inputs + knowledge to decision modules; ensure explainability. • *T9. Develop Human–Machine Interaction Application* Create interface for operators to inspect decisions, provide feedback, and validate reasoning. • *T10. Test and Validate with SME (Human-in-the-loop)* Run test scenarios, collect performance metrics, refine system based on operational feedback. • *T11. Improve and Adapt Algorithm* Enhance rules, enrich KG, optimize inference, and increase decision speed/accuracy. *Milestones:* (M4) Initial Knowledge Graph + rule base completed, (M5) Hybrid AI reasoner prototype operational, (M6) HMI connected and functional, (M7) Validation with SME completed (KPIs measured), (M8) Final improved demonstrator delivered **Methodologies***:* Knowledge Representation (RDF/OWL, ontologies, DPP structures); Semantic Reasoning & Rule-based Logic; Hybrid AI (symbolic + learning components); Multimodal Data Integration (vision, text, lifecycle metadata); Human-in-the-loop decision validation; Industry 4.0 digitalisation practices

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO will support the project with focused technical and research expertise, contributing to the design, development, and validation of the Hybrid AI system for remanufacturing. Key responsibilities include: • **Process Analysis & Formalization** Analyze SME remanufacturing processes and formalize them in structured, machine-readable models. • **AI System Development** Design and implement the core Hybrid AI components, including reasoning logic and integration pipelines. • **Knowledge Representation & Semantic Modelling** Develop and maintain ontologies, Knowledge Graphs, and rule sets for product and process specifications. • **Operational Semantic Systems** Build operational knowledge structures (graphs, rules, DPP alignment) that can be executed in real decision-making scenarios. • **Human/Written Knowledge Extraction** Translate expert knowledge and documentation into formal semantic rules. • **Enterprise Interoperability & Integration** Ensure the AI solution integrates with SME systems, DPP sources, and Industry 4.0 data environments.
Expected R&R of Korean SME Partner	The Korean SME will act as the industrial co-development partner, contributing operational expertise and actively participating in the creation and validation of the Hybrid AI solution. Key responsibilities include: • **Co-define demonstrators and requirements** providing detailed requirement and functional specifications. • **Provide test products and a realistic inspection environment** for data capture, identification, and evaluation. • **Share manufacturing process knowledge and decision rules** to support formalization of product/process logic. • **Co-develop system components** and prototypes together with TNO, contributing engineering effort and participating in joint design and implementation tasks. • **Connect shopfloor data and inspection systems** to the AI solution through engineering support and integration activities. • **Conduct testing and validation** in real operational settings and provide feedback for system refinement.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Currently many manufacturing products (e.g. electronics) are either being fixed manually, recycled, or in the worst-case scenario directly discarded. Manual fixing of discrete products (e.g. electronic PCBs), however, is a time and labor-intensive operation, which often has a disbalance between value of work and value of resulting component, forcing the industries to decide to either discard or recycle it. The purpose of the Hybrid AI Re-X Decision support system is to make remanufacturing faster, more accurate, more informed, and stimulating circularity. It would enable: • Faster and more consistent decision making, reducing labor time and improving throughput, by providing automated decisions about the next product process step based on pre-defined rules. • Better-informed decisions, supported by richer product and process information, achieved by introducing multi-modal product information (e.g. DPP data, BOM, Inspection Visuals, Analysis) • Higher circularity, as improved insight and faster decisions increase the number of products that can be repaired, reused, or repurposed. • Lower operation costs and a stronger business case for SME remanufacturing, as automated decision process makes it faster, cheaper and easier to process components. • Critical knowledge captures in a machine-readable way, which allows SME to have a clear overview of the way decisions are taken. The added traceability of the automated decisions would enable also analysis and check into why a choice has been made and what impact

	that has had on the next steps. This also allows for building next SI applications, which can replace manual steps. • Reduced Training Time: Employees are supported by guided, system-generated actions reducing time to master the vast expertise from extensive manuals and tacit knowledge.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization